

LINEAR ALGEBRA — ASSIGNMENT 1.

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CS2604.

1. True / False?

a) The intersection of two subspaces, is always a subspace.

- True.

Proof:- Let's say we have two subspaces U, V

we have to show $U \cap V$ is also a subspace.

We know, any subspace must satisfy, these three given properties,

- Contains zero vector :- Any subspace, X must contain $\vec{0}$, i.e. $\vec{0} \in X$.

- Closed under ^{vector} addition & ^{scalar} multiplication :- Any vector $\vec{a}, \vec{b} \in X, c \in \mathbb{F}$.

Then, $c\vec{a} + \vec{b} = \vec{v} \in X$.

Consider our subspaces U and V .

i) As U and V are subspaces, they must contain $\vec{0}$,

i.e. $\vec{0} \in U$ and $\vec{0} \in V$

So, togetherly, we can write $\vec{0} \in U \cap V$. (So $U \cap V$ contains $\vec{0}$)

ii) Let us assume two vectors $\vec{x}, \vec{y} \in U \cap V$.

So, $\vec{x}, \vec{y} \in U$, U is a subspace (closed under addition) $\Rightarrow \vec{x} + \vec{y} \in U$ — (i)

and $\vec{x}, \vec{y} \in V$, V is a subspace (closed under addition) $\Rightarrow \vec{x} + \vec{y} \in V$ — (ii)

from (i) and (ii) we can conclude, $\vec{x} + \vec{y} \in U \cap V$. (So $U \cap V$ is closed under addition).

iii) Let us take another vector $\vec{a} \in U \cap V$, $c \in \mathbb{F}$.

So, $\vec{a} \in U$, U is a subspace (closed under sc. mult.) $\Rightarrow c \cdot \vec{a} \in U$ — (i)

and, $\vec{a} \in V$, V is a subspace (closed under sc. mult.) $\Rightarrow c \cdot \vec{a} \in V$ — (ii).

from (i) & (ii) we can conclude, $c\vec{a} \in U \cap V$. (So $U \cap V$ is also closed under sc. multiplication).

So we showed $U \cap V$ is satisfying all the three properties of a sub-space,
So, $U \cap V$ is a subspace.

b) The union of two sub space is always a subspace.

False.

Citing as counter-example.

Let $\mathbb{R}^3$ be our vector-space and $U \in \mathbb{R}^2$ and $W \in \mathbb{R}^2$ be two
valid subspaces. defined as,

$$U = \{(x, y, 0) \mid x, y \in \mathbb{R}\} \text{ and } W = \{(0, y, z) \mid y, z \in \mathbb{R}\}.$$

So, vector $(1, 2, 0) \in U$ and

$$(0, 2, 3) \in W$$

union of these,

$$(1, 4, 3) \notin U \text{ and } (1, 4, 3) \notin W.$$

$$\therefore (1, 4, 3) \notin U \cup W.$$

So, union of two subspace is not a subspace.

c) The sum of two subspace below, determine a subspace.

True

We know, Sum of two subspace is defined as,

$$U + V = \{\bar{a} + \bar{b} \mid \bar{a} \in U \text{ and } \bar{b} \in V\} \text{ i.e. } (\bar{a} + \bar{b}) \in U + V.$$

Proof, $U + V$ is also a sub-space, if U, V are sub space
given.

1. As U, V are subspace, $\bar{0} \in U, \bar{0} \in V$.

$$\text{So, } U + V = \{\bar{0} + \bar{0} = \bar{0}\} ; \boxed{\bar{0} \in U + V}$$

2. Let $\bar{x}, \bar{y} \in U + V$.

By definition, $\bar{x} = \bar{x}_u + \bar{x}_v$ and $\bar{y} = \bar{y}_u + \bar{y}_v$.

where $\bar{x}_u, \bar{y}_u \in U$ and $\bar{x}_v, \bar{y}_v \in V$.

If we add $\bar{x} + \bar{y}$ we get

$$= (\bar{x}_u + \bar{x}_v) + (\bar{y}_u + \bar{y}_v)$$

$$= (\bar{x}_u + \bar{y}_u) + (\bar{x}_v + \bar{y}_v)$$

We know, $\bar{x}_u + \bar{y}_u \in U \Rightarrow \bar{x}_u + \bar{y}_u \in x_u + u = u$

Similarly $\bar{x}_v + \bar{y}_v \in V \Rightarrow \bar{x}_v + \bar{y}_v \in v + v = v$

$$= (\bar{x}_u + \bar{y}_u) + (\bar{x}_v + \bar{y}_v) \in u + v.$$

$$\Rightarrow \boxed{\bar{x} + \bar{y} \in u + v.}$$

3. Let $\bar{a} \in u + v$, so, $\bar{a}$ can be written as $\bar{a} = \bar{a}_u + \bar{a}_v$ where $\bar{a}_u \in u$ and $\bar{a}_v \in v$.

As, u, v is a subspace, on field $\mathbb{F}$

we can scale, the vectors by $c \in \mathbb{F}$

$$c\bar{a}_u \in u \text{ and } c\bar{a}_v \in v.$$

$$\Rightarrow c\bar{a}_u + c\bar{a}_v \in u + v.$$

$$\Rightarrow c(\bar{a}_u + \bar{a}_v) \in u + v \Rightarrow \boxed{c\bar{a} \in u + v.}$$

d) Every vector space is the direct sum of two proper subspace.

False.

Citing an counter-example.

Let's say our vector space is $\mathbb{R}^1$ - ie $V = \mathbb{R}^1$

$\mathbb{R}^1$ only has two sub-space, $\mathbb{R}^1$ itself and null space $\{0\}$.

And only one proper sub-space $\{0\}$.

We can't make $\mathbb{R}^1$ from direct sum of $\{0\}$

So our given statement fails.

e) If $U \cap W = \{0\}$, then $U + W = V$.

False.

Let's say $V = \mathbb{R}^3$ and $U = \{(x, 0, 0) \mid x \in \mathbb{R}\}$ and $W = \{(0, 0, z) \mid z \in \mathbb{R}\}$.

with the following U and W we have $U \cap W = \{0\}$.

As U is the x axis and W is the z axis, both only have the origin at common.

If we add $U + W$ we get $\{(x, 0, z) \mid x, z \in \mathbb{R}\}$ is the xz plane.

So, xz plane $\neq$ the whole $\mathbb{R}^3$ space.

As, $(1, 2, 3) \in \mathbb{R}^3$ but $(1, 2, 3) \notin U + W$.

$\therefore U + W \neq V$.

2. Classify all possibility: For U & W are subspace of V .

I. $U = W$.

a) Geometric picture

Let our vector space V be $\mathbb{R}^2$.

and sub-space U, W be $\mathbb{R}^1$.

ie, $U = W = \{(x, 1), x \in \mathbb{R}\}$.

while the whole 2-D xy plane is $\mathbb{R}^2$, a line passing through $y=1$ parallel to x -axis is our sub-space U/W .

b) If $U \cup W$ is a subspace.

We know, U & W are both sub-spaces individually,

and given $U = W$.

So by set rules, $U \cup W = U/W$ is a sub-space.

So $U \cup W$ is also a sub-space.

c) If $U+W$ is a direct sum?

Recalling the condition of direct sum:-

Given two sub spaces U, W of a vector space V , represented as

$$V = U \oplus W \text{ iff:}$$

i) all vector in V can be written as a sum of vector from U and W ($\vec{v} \in V \Rightarrow \vec{v} = \vec{u} + \vec{w}; \vec{u} \in U \text{ \& } \vec{w} \in W$).

$$\text{ii) } U \cap W = \{\vec{0}\}.$$

In our given condition, $U=W$.

$$\text{So, } U \cap W = U/W \neq \{\vec{0}\}$$

is failing the condition, hence $U+W$ is NOT a direct sum.

II. $U \subsetneq W$

a) Geometric picture.

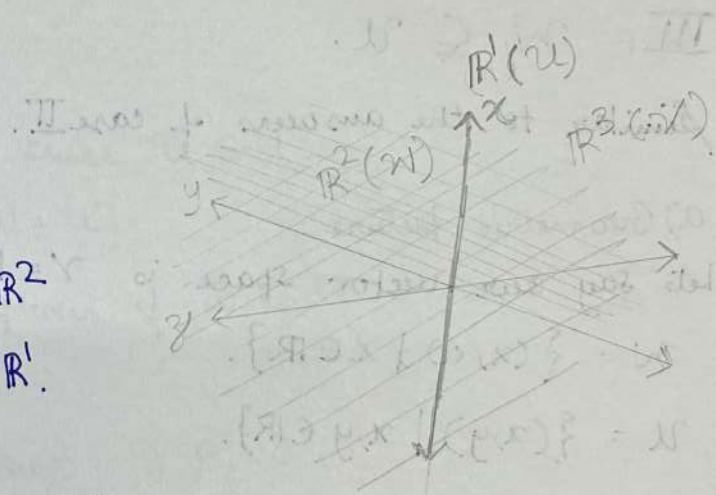
Let our vector space be $\mathbb{R}^3$.

and sub-space W as xy plane is $\mathbb{R}^2$

and sub-space U as x line is $\mathbb{R}^1$.

$$W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}.$$

$$U = \{(x, 0, 0) \mid x \in \mathbb{R}\}.$$



b) If $U \cup W$ a sub space?

We know, U, W are sub space individually.

and U is a proper sub set of W . $U \subsetneq W$.

$$\text{So, } U \cup W = W \text{ (by set rules)}$$

Hence, we conclude $U \cup W$ is a sub space.

c) If $U+W$ is a direct sum?

We know, $U \not\subseteq W$.

for direct sum, i.e. $V = U \oplus W$ we must have, $U \cap W = \{0\}$

But $U \cap W = U \neq \{0\}$.

So, $U+W$ is not a direct sum.

BUT, if we know, $\{0\}$ is also a valid proper subspace. (ii)

If $U = \{0\}$

Then $U \cap W = U = \{0\}$, then $U+W$ is a direct sum.

So $U+W$ could / could NOT be a direct sum depending on subspace definition.

III. $W \subsetneq U$.

Similar to the answers of case II.

a) Geometric picture.

Let's say our vector space is $V = \mathbb{R}^2$.

$$W = \{(x, 0) \mid x \in \mathbb{R}\}.$$

$$U = \{(x, y) \mid x, y \in \mathbb{R}\}.$$

W is the x axis and U is the whole xy plane.

b) If $U \cup W$ is a subspace?

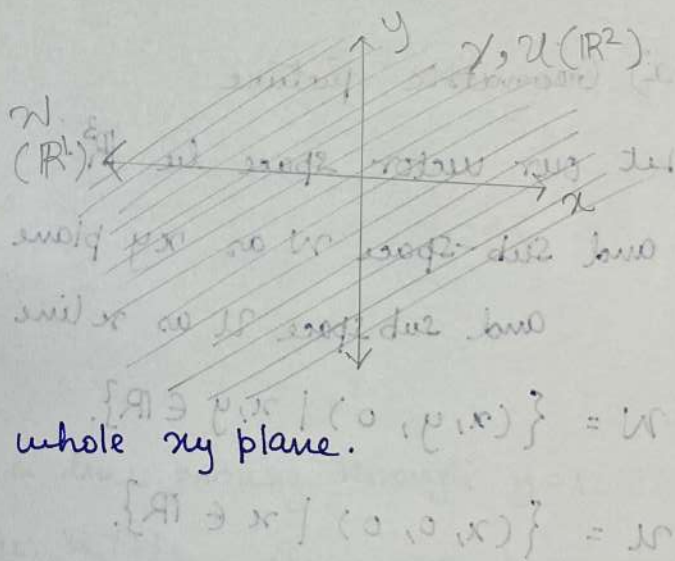
As we have, $W \subsetneq U$, $U \cup W = U$.

and U, W both are subspaces, $U \cup W$ is also a subspace.

c) If $U+W$ is a direct sum?

CASE I, if $W = \{0\} \Rightarrow U \cap W = W = \{0\}$ is direct sum.

CASE II, else $W \neq \{0\} \Rightarrow U \cap W = W \neq \{0\}$ is NOT direct sum.



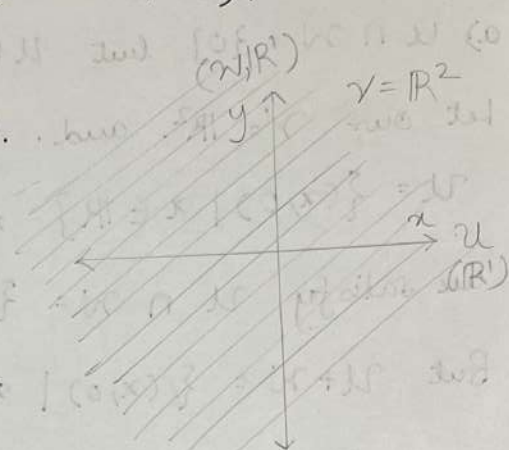
IV. Neither contains the other. ($U \not\subseteq W$ & $W \not\subseteq U$).

a) Geometric picture.

Let's consider our vector space $V = \mathbb{R}^2$.

And $U = \{(x, 0) \mid x \in \mathbb{R}\}$ is x -axis

$W = \{(0, y) \mid y \in \mathbb{R}\}$ is y -axis



b) If $U \cup W$ is a subspace.

No, we can't find a subspace from two such subspaces where nothing is contained inside each other.

We can find a vector $\bar{u} \in U$ (but $\bar{u} \notin W$) and $\bar{w} \in W$ (but $\bar{w} \notin U$) such that $\bar{u} + \bar{w}$ falls outside $U \cup W$.

Let's assume $U \cup W$ is a subspace.

Then $\bar{u} + \bar{w} \in U \cup W$.

Let's assume, $\bar{u} + \bar{w} \in U$ and since $\bar{u} \in U$.

$\bar{w} = (\bar{u} + \bar{w}) - \bar{u} \in U$. [$U + U = U$].

But this contradicts our assumption, of $\bar{w} \notin U$.

Similarly if $\bar{u} + \bar{w} \in W$.

We can prove via contradiction that $\bar{u} \notin W$.

So, $U \cup W$ is NOT a subspace.

c) If $U + W$ is a direct sum?

If our vector space is $\mathbb{R}^2$ and U and W be x and y axis respectively

then $U \cap W = \{\vec{0}\}$ is direct sum.

If our vector space is $\mathbb{R}^3$ and U and W be xy and yz plane

$U \cap W = y \text{ axis} \neq \{\vec{0}\}$ is not direct sum

So it will depend on our choice of U, W .

3. Construct Examples.

a) $U \cap W = \{0\}$ but $U + W \neq V$.

Let our $V = \mathbb{R}^2$ and.

$$U = \{(x, 0) \mid x \in \mathbb{R}\}, \quad W = \{(0, 0)\}.$$

We satisfy $U \cap W = \{0\}$ and U, W are valid sub-spaces of V .

But $U + W = \{(x, 0) \mid x \in \mathbb{R}\} \neq \mathbb{R}^2$.

b) $U + W = V$ but $U \cap W \neq \{0\}$.

Let our $V = \mathbb{R}^2$ and.

$$U = \{(x, y) \mid x, y \in \mathbb{R}\}, \quad W = \{(x, 0) \mid x \in \mathbb{R}\}.$$

$$U + W = \{(x, y) \mid x, y \in \mathbb{R}\} \text{ is } \mathbb{R}^2.$$

But $U \cap W = \{(x, 0) \mid x \in \mathbb{R}\}$ - i.e. x -axis.

c) $V = U \oplus W$.

Let our $V = \mathbb{R}^2$ and

$$U = \{(x, 0) \mid x \in \mathbb{R}\}, \quad W = \{(0, y) \mid y \in \mathbb{R}\}.$$

$V = U \oplus W$ is satisfied as $U \cap W = \{0\}$ and.

$$U + W = V. \text{ i.e. } V = \{(x, y) \mid x, y \in \mathbb{R}\}, \mathbb{R}^2.$$

d) $U = W$.

Let our $V = \mathbb{R}^2$ and.

$$U = \{(x, x) \mid x \in \mathbb{R}\}, \quad W = \{(x, y) \mid x = y \text{ \& } x, y \in \mathbb{R}\}.$$

4. The Converse Game.

a) If $V = U \oplus W$, then $U \cap W = \{0\}$.

CONVERSE: $U \cap W = \{0\} \Rightarrow V = U \oplus W$.

False.

As we have shown, in Q3a,

If our vector space is $V = \mathbb{R}^2$,

$$U = \{(x, 0) \mid x \in \mathbb{R}\} \text{ and } W = \{(0, 0)\}.$$

we have $U \cap W = \{0\}$, but $U \oplus W = \{(x, 0) \mid x \in \mathbb{R}\} \neq \mathbb{R}^2$.

b) If $V = U \oplus W$, then every vector in V has a unique decomposition.

CONVERSE: If V has a unique decomposition $\rightarrow V = U \oplus W$.

TRUE

Proof: By definition, $V = U \oplus W$ requires two conditions:

$$V = U + W \text{ and } U \cap W = \{0\}.$$

1. Prove $U + W = V$.

The given statements premise explicitly states that every $\vec{v} \in V$ can be written as $u + w$. Therefore, the sum generates the entire space, so $V = U + W$.

2. Prove $U \cap W = \{0\}$.

Let $x \in U \cap W$. We can express the zero vector $\vec{0} \in V$ in two ways:

$$\hookrightarrow \vec{0} = \vec{0} + \vec{0} \text{ \{where } 0 \in U \wedge 0 \in W\}}$$

$$\hookrightarrow \vec{0} = x + (-x) \text{ \{where } x \in U \wedge -x \in W\}}$$

Because the premise guarantees that every vector's decomposition is strictly unique, the two exp are identical.

$$\text{Hence } U \cap W = \{0\} \Rightarrow V = U \oplus W$$

c) If $U + W = V$ then $V = U \oplus W$.

CONVERSE : If $V = U \oplus W$ then $U + W = V$.

The converse is true.

As if we are given $V = U \oplus W$ that means

we have also indirectly given,

i. $V = U + W$ and

ii. $U \cap W = \{0\}$.

Hence via statement (i) we conclude $V = U + W$ is true.

BUT the original statement, If $U + W = V \rightarrow V = U \oplus W$ is

false.

Citing an counter-example,

Let $V = \mathbb{R}^2$. $U = \{(x, y) | x, y \in \mathbb{R}\}$; $W = \{(0, y) | y \in \mathbb{R}\}$.

we have $U + W = V$ but $U \cap W \neq \{(0, y) | y \in \mathbb{R}\} \neq \{0\}$.

ie $U + W = V$ alone can't guarantee $V = U \oplus W$.

5. A characterization of Direct Sum.

Given :- A vector space V and two of its subspaces U and W .

To Prove :- $U + W$ is a direct Sum iff

$$U + W = 0 \text{ implies } U = 0, W = 0.$$

or in notations,

$$U + W \text{ is direct} \iff (U + W = 0 \Rightarrow U = 0, W = 0).$$

As we have a iff sign ie ($\Leftrightarrow$) let's say $A \Leftrightarrow B$
we need to prove $A \Rightarrow B$ and $B \Rightarrow A$.

1. FORWARD PROOF.

Given : $U+W$ is a direct Sum.

To Prove : $\bar{u} + \bar{w} = 0 \Rightarrow \bar{u} = 0$ and $\bar{w} = 0$.

Proof :- As $U+W$ is a direct Sum.

we are given / know two condition :-

• $U+W = V$ and $U \cap W = \{0\}$.

let us assume two vectors $\bar{u} \in U$ and $\bar{w} \in W$

Such that $\bar{u} + \bar{w} = 0$

$\Rightarrow \bar{u} = -\bar{w}$

we know, $\bar{w} \in W$, and as W is a subspace; $-\bar{w} \in W$.

again, $\bar{u} \in U$.

So, $\bar{u} \in U$ and $\bar{u} \in W \Rightarrow \bar{u} \in U \cap W$.

But we know, $U \cap W = \{0\}$.

Then $\bar{u} = \{0\}$.

$\Rightarrow \bar{u} + \bar{w} = 0$

$\Rightarrow 0 + \bar{w} = 0 \Rightarrow \bar{w} = \{0\}$.

Hence we showed if $U+W$ is direct Sum then

$\bar{u} + \bar{w} = 0 \Rightarrow \bar{u} = 0 \wedge \bar{w} = 0$.

2. BACKWARD PROOF: let $u \in U$ and $w \in W$ such that $u+w=0$.

Given:- For any $\bar{u} \in U$ and $\bar{w} \in W$, we have $\bar{u} + \bar{w} = 0 \Rightarrow$
 $\bar{u} = 0 \wedge \bar{w} = 0$.

To Prove:- $U + W$ is a direct sum. (i.e. $U \cap W = \{0\}$).

Proof:- Let $x \in U \cap W$.
 $\Rightarrow x \in U \wedge x \in W$.

As W is a subspace, $-x \in W$.

Adding, we get $x + (-x) = 0$.

Comparing with our assumption $\bar{u} + \bar{w} = 0$.

we get $\bar{u} = x$ and $\bar{w} = -x$.

But we know $\bar{u} = 0 \wedge \bar{w} = 0$.

Hence, $x(=0)$.

$\therefore$ Arbitrary vector is $0 \in U \cap W$.

So, $U \cap W = \{0\} \Rightarrow U + W$ is direct sum.

So, we proved both direction of bi-implication,

So, our proof is complete.

6. Three Subspaces.

Find subspaces U , W and Z of $\mathbb{R}^3$ such that,

$$\mathbb{R}^3 = U \oplus W \oplus Z.$$

Write $(5, -2, 7)$ as sum of U , W and Z .

Can we find another triple of sub space with same property?

To make $\mathbb{R}^3$ the direct sum of three sub-spaces, the simplest approach is we can use the standard basis for $\mathbb{R}^3$.

$$\left. \begin{aligned} U &= \{(x, 0, 0) \mid x \in \mathbb{R}\}, \\ W &= \{(0, y, 0) \mid y \in \mathbb{R}\}, \\ Z &= \{(0, 0, z) \mid z \in \mathbb{R}\}. \end{aligned} \right\} \begin{aligned} &\text{we can write} \\ &\mathbb{R}^3 = U \oplus W \oplus Z. \\ &\text{As, } U \cap W \cap Z = \{0\}. \end{aligned}$$

and

$$U + W + Z = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}.$$

So, we can write vector $(5, -2, 7)$ as a sum of U , W , Z as,

$$(5, -2, 7) = \underbrace{(5, 0, 0)}_{\in U} + \underbrace{(0, -2, 0)}_{\in W} + \underbrace{(0, 0, 7)}_{\in Z}.$$

Another Basis,

If we consider three lines on $\mathbb{R}^3$, and intersect only in point origin i.e. $(0, 0, 0)$ and the three vectors are linearly independent of each other, i.e. they can span the whole $\mathbb{R}^3$ space.

Let U_{new} be a line $y=x$; $U_{\text{new}} = \{(a, a, 0) \mid a \in \mathbb{R}\}.$

Let W_{new} be another line $y=-x$; $W_{\text{new}} = \{(b, -b, 0) \mid b \in \mathbb{R}\}.$

Let Z_{new} be Z -axis, i.e. $Z_{\text{new}} = \{(0, 0, c) \mid c \in \mathbb{R}\}.$

As these lines intersect only at origin and.

$$\mathbb{R}^3 = \underset{\text{new}}{U} + \underset{\text{new}}{W} + \underset{\text{new}}{Z} = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}.$$

Decomposing $(5, -2, 7)$ with new triple,

$$(a, a, 0) + (b, -b, 0) + (0, 0, c) = (5, -2, 7).$$

So,

$$a + b = 5$$

$$a - b = -2$$

$$c = 7$$

$$\hline 2a = 3$$

$$a = 1.5, \quad b = 3.5, \quad c = 7.$$

$\therefore$ The vectors are,

$$\text{New} = \{(1.5, 1.5, 0), (3.5, -3.5, 0), (0, 0, 7)\}.$$

This is the unique representation of $(5, -2, 7)$.

8. A Beautiful Theorem.

Suppose U and W are subspaces of V .

To prove:- $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Proof :- BACKWARD DIRECTION.

Assuming $U \subseteq W$ OR $W \subseteq U$.

$\hookrightarrow$ If $U \subseteq W$, then $U \cup W = W$. Since W is given as a subspace,

So $U \cup W$ is a subspace.

$\hookrightarrow$ If $W \subseteq U$, then $U \cup W = U$. Since U is given as a subspace,

$U \cup W$ is a subspace.

Thus in either case, the union is a subspace.

FORWARD DIRECTION.

Assuming $U \cup W$ is a subspace.

Assume $U \subsetneq W$ and $W \subsetneq U$.

- So, as $U \subsetneq W$, there exists a vector $\bar{u} \in U$ which $\bar{u} \notin W$.
- Again as $W \subsetneq U$, there exists a vector $\bar{w} \in W$ which $\bar{w} \notin U$.
- Since $\bar{u} \in U$ and $\bar{w} \in W$, both $\bar{u}, \bar{w} \in U \cup W$.
- Again we assumed $U \cup W$ is a subspace, it is closed under addition.

So, a vector $\bar{v} = \bar{u} + \bar{w}$, $\bar{v} \in U \cup W$.

- If $\bar{v} \in U \cup W$, then either $\bar{v} \in U$ OR $\bar{v} \in W$.

↳ Let's say $\bar{v} \in U$: Since $\bar{u} \in U$ and U is a subspace, then $-\bar{u} \in U$. Then $\bar{v} - \bar{u} = (\bar{u} + \bar{w}) - \bar{u} = \bar{w}$. This implies $\bar{w} \in U$, which contradicts our assumption of $\bar{w} \notin U$.

↳ Let's say $\bar{v} \in W$: Since $\bar{w} \in W$ and W is a subspace, $-\bar{w} \in W$. Then $\bar{v} - \bar{w} = (\bar{u} + \bar{w}) - \bar{w} = \bar{u}$.

This implies $\bar{u} \in W$, which contradicts our initial $\bar{u} \notin W$,
Setup

∴ Both cases lead to a contradiction, our assumption, must be false.

Therefore, either $U \subseteq W$ OR $W \subseteq U$.

10. A New Operation?

Suppose we define: $U \circledast W = \{u - w : u \in U, w \in W\}$

Is this genuinely a new operation, or is it just sum of two subspaces?

Justify?

Proof: We know, the sum of two subspaces is $U + W =$

$$\{u + w : u \in U, w \in W\}.$$

To show $U \circledast W \cong U + W$, we must show that each set is a subset of the other.

↳ To show $U \circledast W \subseteq U + W$.

We know, $U \circledast W = \{u - w \text{ for some } u \in U \text{ and } w \in W\}$.

W is a subspace, is closed under scalar multiplication.

$$\therefore -w \in W.$$

$$\text{rewriting } u - w = u + (-w) = \bar{v}.$$

Since, $u \in U$ and $-w \in W$, by definition of standard sum,

$$\bar{v} \in U + W.$$

↳ To show $U + W \subseteq U \circledast W$.

Let $\bar{v} \in U + W$. By definition, $\bar{v} = u + w$ for some $u \in U$ and $w \in W$.

W is a subspace, is closed under scalar multiplication, and $-w \in W$.

$$\text{Let } -w = x. \quad x = -w.$$

$$\text{So, } \bar{v} = u + w = u - (-w) = u - (x).$$

$$\bar{v} = \bar{u} - \bar{x}$$

Since $u \in U$ and $x \in W$, this expression $u - x$ perfectly matches the definition of new operation. $\therefore \bar{v} \in U \oplus W$.

So, we showed,

$$U \oplus W \subseteq U + V$$

$$U + V \subseteq U \oplus W$$

So we conclude,

$$U \oplus W = U + V$$

9. Think like a mathematician.

Give three proofs that

$\{0\}$ is a subspace of every vector space.

Which proof do we think to be most elegant? Explain.

PROOF 1:-

Let's assume $\{0\}$ is a subspace of every vector space.

Checking:-

i) Contains the Zero Vector:- As we are checking for $\{0\}$,
 $0 \in \{0\}$

ii) Closed under addition :- As $0 \in \{0\}$. And $0 + 0 = 0 \in \{0\}$.

iii) Closed under multiplication :- $c \in \mathbb{F}$; $c0 = 0 \in \{0\}$.

As we checked all condition for subspace, hence $\{0\}$ is one.

PROOF 2:-

Let's take a vector $\vec{v} = \vec{0}$

Taking $\text{span}(\vec{v})$ is all linear combination of $\vec{v}$.

ie $c \cdot \vec{v}$ where $c \in \mathbb{F}$.

But $\vec{v} = \vec{0} \Rightarrow c \cdot \vec{v} = \vec{0}$.

$\therefore \text{span}\{\vec{0}\} = \{\vec{0}\}$.

Since all spans are valid sub-spaces, $\{\vec{0}\}$ is a subspace.

PROOF 3:-

We know, intersection of n subspaces is always a subspace.

And we know, all subspace must contain $\{\vec{0}\}$

So intersection of all subspaces must contain $\{\vec{0}\}$ and also form a subspace.

According to me,

The 'proof 1' is best, as we checked our three condition of a subspace and then conclude that it is $\{\vec{0}\}$ is a sub-space.

7. Generalization

Suppose $V = U_1 \oplus U_2 \oplus \dots \oplus U_n$.

Guess what condition replaces $U \cap W = \{0\}$.

in the definition of a direct sum of n subspaces.

State the conjecture and prove it for the case of $n=3$.

Given,

$$V = U_1 \oplus U_2 \oplus \dots \oplus U_n \text{ which means,}$$

$$\hookrightarrow V = U_1 + U_2 + \dots + U_n \text{ and for}$$

$$\hookrightarrow \text{for every } i, U_i \cap \left(\sum_{j \neq i} U_j \right) = \{0\}.$$

Also,

$$U_1 + U_2 + \dots + U_n = 0, \quad U_i \in U_i \Rightarrow U_1 = U_2 = \dots = U_n = 0.$$

The Proof (For $n=3$).

We must prove that $V = U_1 + U_2 + U_3$ is a direct sum, i.e.

$$1. U_1 \cap (U_2 + U_3) = \{0\}.$$

$$2. U_2 \cap (U_1 + U_3) = \{0\}.$$

$$3. U_3 \cap (U_1 + U_2) = \{0\}.$$

FORWARD DIRECTION:

Assume $V = U_1 \oplus U_2 \oplus U_3$ is a direct sum.

So, $V = U_1 + U_2 + U_3$ is unique by definition.

1. Let $\bar{x}$ be an arbitrary vector in the intersection $U_1 \cap (U_2 + U_3)$.
2. Because it is in the intersection, $x \in U_1$ and we can also write $\bar{x} = u_2 + u_3$ for some $u_2 \in U_2$ & $u_3 \in U_3$.
3. Rearranging: $x - u_2 - u_3 = 0$.
4. $\therefore 0 = 0 + 0 + 0$ (trivially) where $0 \in U_1, U_2, U_3$.
Our premise guarantees that decompositions are unique, our equation must exactly match the trivial one.
5. $\therefore$ the U_1 component must be zero ($x=0$), the U_2 component must be zero ($-u_2=0$), and the U_3 component must be zero ($-u_3=0$).
6. Since $x=0$, the intersection $U_1 \cap (U_2 + U_3)$ is strictly $\{0\}$.

BACKWARD DIRECTION :

Assume the three intersection conditions are strictly equal to $\{0\}$.
We must prove that any vector $\bar{v} \in V$ has a unique decomposition

1. Suppose a vector $\bar{v}$ has two decompositions:-
$$v = \bar{x}_1 + \bar{x}_2 + \bar{x}_3 \quad \& \quad v = y_1 + y_2 + y_3; \text{ where } x_i, y_i \in U_i.$$

By rearranging the terms; $x_1 - y_1 = (y_2 - x_2) + (y_3 - x_3)$.

Notice that the left side of eq is U_1 i.e. $(x_1 - y_1) \in U_1$.

Right side belongs to $(y_2 - x_2) + (y_3 - x_3) \in U_2 + U_3$.

5. Because the LHS = RHS, this vector must belong in U_1 and $(U_2 + U_3)$. i.e., in intersection $U_1 \cap (U_2 + U_3)$.

6. By our initial assumption, this intersection is $\{0\}$.

$$\therefore x_1 - y_1 = 0 \Rightarrow x_1 = y_1.$$

7. Now, substitute $x_1 = y_1$ back into our equation from step 1.

This cancels our first term, leaving us with:

$$x_2 + x_3 = y_2 + y_3 \Rightarrow x_2 - y_2 = y_3 - x_3.$$

8. Here the left side $\in U_2$, RHS $\in U_3$. i.e. They belong to $U_2 \cap U_3$.

9. Because U_3 is a subset of $U_1 + U_3$, the intersection of just U_2 and U_3 is even smaller than $U_2 \cap (U_1 + U_3)$, which we assumed to be $\{0\}$. $\therefore U_2 \cap U_3 = \{0\}$.

10. This means :

$$x_2 - y_2 = 0 \Rightarrow x_2 = y_2.$$

$$y_3 - x_3 = 0 \Rightarrow x_3 = y_3.$$

11. As, $x_1 = y_1$, $x_2 = y_2$ and $x_3 = y_3$, the decomposition is perfectly unique. $U_1 + U_2 + U_3$ is a direct sum.